

Complaint Management and Tracking System

(CMTS)

Database Design Document

V 1.0

By

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15/05/26	V 1.0	Initial submission of the CMTS Database Design Document, covering entity identification, data dictionary, business rules, and EER diagram.	—

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION

The Complaint Management and Tracking System (CMTS) is a platform designed to digitalize complaint processing within a university. It ensures that every campus facility issue is reported, tracked, and resolved efficiently. The proposed system will serve multiple types of users including students, faculty, staff members, and administration [1]. Students can report any issue immediately and see the progress. Faculty and staff can also file complaints regarding workplace issues as well as play a role in resolving students' complaints. Administration can submit complaints, resolve them, update their status, and generate performance reports.

The purpose of CMTS is to replace ad-hoc, manual complaint handling via email, paper forms, or verbal communication with a structured, role-based, and fully audit-able digital workflow [2]. The system supports the complete life cycle of a complaint: submission, categorization, assignment, status tracking, resolution, and post-resolution feedback.

1.2. PROBLEM STATEMENT

In the manual complaint handling system within a university, complaints are submitted via emails or verbal communication. There is no system to record complaints and track them to check progress [1]. Communication between complainants and the administration is limited, which delays response time and complaint resolution. Complainants cannot see any progress on their complaint. This unstructured system causes inefficiency and decreases user satisfaction.

The CMTS solves this problem through a digital platform which enables complainants to submit their complaints [2]. It enables them to track their complaint and receive status updates by assigning a unique ID to each complaint. It aims to enhance response and resolution times by keeping a record of all registered complaints. It allows administrators to generate reports and analyze them for future improvements.

1.3. PROJECT OBJECTIVES

General Objective: To design and implement a centralised, role-based complaint management and tracking system that enables all Namal University stakeholders to submit, monitor, and resolve issues efficiently and transparently [1].

Specific Objectives:

- To reduce response time and resolution delays caused by manual management of complaints [1].
- To enable complaint tracking where users receive updates throughout the resolution process [1].
- To enhance efficiency by allowing administrators to analyze performance and generate reports [2].
- To increase user satisfaction through feedback mechanisms and improve transparency in handling complaints [1].

1.4. GITHUB REPOSITORY LINK:

Click here to view the repository

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY

The following entities form the core of the CMTS relational schema.

Sr. No	Entity Name	Description
01	User (Supertype)	Every person who can authenticate into CMTS. Stores shared identity and authentication attributes [1].
02	Student	Subtype of User representing undergraduate students who are primary complainants [1].
03	Faculty	Subtype of User representing teaching staff [1].
04	Staff	Subtype of User representing administrative and technical personnel [1].
05	Administrator	Subtype of User representing system-wide operators with full access [1].
06	Complaint_Handler	Subtype of User representing departmental complaint handlers assigned to specific trackers [1].
07	Department	University departments (e.g., Academics, IT, Finance) to which complaints are routed [1].
08	Tracker	Complaint categories within a department (e.g., Exam Scheduling under Academics) used to classify and route submissions [1].
09	Complaint	Central entity representing a single complaint from submission through resolution [1].
10	Status_History	Audit trail of every status transition on a complaint [1].
11	Attachment	Metadata for image files uploaded as supporting evidence with a complaint [1].
12	Comment	Text messages posted by complainants or handlers on a complaint [1].
13	Feedback	Post-resolution satisfaction rating submitted by the original complainant [1].
14	Audit_Log	Log of all critical system actions for security and accountability [1].

Table 2.1: CMTS Entity Identification

2.2. DATA DICTIONARY

For each entity, all attributes are defined with their data type, constraints, and a description of their role within the system. Foreign keys are not included in the data dictionary. They are documented in the relationship section.

2.2.1. User (Supertype)

SRS References: 3.1.1.1, 2.3.1–2.3.4, 3.4.1.1, 3.4.1.3 [1]

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	NOT NULL, AUTO_INCREMENT	System-generated unique identifier.
02	full_name	VARCHAR(150)	NOT NULL	Displayed on complaints and comments for identification.
03	email	VARCHAR(255)	NOT NULL, UNIQUE	University email used as login credential.
04	password_hash	VARCHAR(255)	NOT NULL	Hashed password for secure credential verification [1].
05	role	ENUM	NOT NULL (Student Faculty Staff Administrator)	Determines database permissions and permitted operations.
06	account_status	ENUM	NOT NULL (Active Locked Inactive)	Supports account lock-out after five failed logins per BR-02. Admins can unlock.
07	failed_login_count	TINYINT	NOT NULL, DEFAULT 0	Counter incremented on each failed login; triggers lockout at 5 per BR-02 [1].
08	device_token	VARCHAR(512)	NULLABLE	Firebase Cloud Messaging token for push notifications to the user's device.
09	created_at	TIMESTAMP	NOT NULL	Account creation timestamp required for audit reporting [1].
10	updated_at	TIMESTAMP	NOT NULL	Tracks the last profile modification for data integrity.
11	user_type	ENUM	NOT NULL	Stores the primary subtype a user belongs to.
12	is_student	BOOLEAN	NOT NULL, DEFAULT FALSE	Identifies whether the user belongs to the Student subtype.
13	is_faculty	BOOLEAN	NOT NULL, DEFAULT FALSE	Identifies whether the user belongs to the Faculty subtype.
14	is_staff	BOOLEAN	NOT NULL, DEFAULT FALSE	Identifies whether the user belongs to the Staff subtype.

#	Name	Data Type	Constraint	Description
15	is_administrator	BOOLEAN	NOT NULL, DEFAULT FALSE	Identifies whether the user holds the Administrator functional role.
16	is_complaint_handler	BOOLEAN	NOT NULL, DEFAULT FALSE	Identifies whether the user holds the Complaint_Handler functional role.

Table 2.2: User Entity Attributes

2.2.2. Student (Subtype of User)

SRS References: 2.3.1 [1]

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	PK, NOT NULL	Shared PK with User enforces the supertype–subtype relationship.
02	reg_number	VARCHAR(30)	NOT NULL, UNIQUE	University registration number referenced on complaint records.
03	program	VARCHAR(100)	NOT NULL	Degree programme (e.g., BS Computer Science) for routing.
04	enrollment_date	DATE	NOT NULL	Date of enrollment used for analytics and cohort-based complaint reporting.

Table 2.3: Student Entity Attributes

2.2.3. Faculty (Subtype of User)

SRS References: 2.3.2 [1]

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	PK, NOT NULL	Shared PK enforces supertype–subtype relationship.
02	designation	VARCHAR(100)	NOT NULL	Job title (e.g., Lecturer, Assistant Professor) for identification.
03	specialization	VARCHAR(150)	NULLABLE	Subject area; supports analytics on faculty complaint patterns.
04	office_number	VARCHAR(10)	NULLABLE	Office room number; useful when a complaint references a specific office location.

#	Name	Data Type	Constraint	Description
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Table 2.4: Faculty Entity Attributes

2.2.4. Staff (Subtype of User)

SRS References: 2.3.3 [1]

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	PK, NOT NULL	Shared PK enforces supertype–subtype relationship .
02	position	VARCHAR(100)	NOT NULL	Job position (e.g., Lab Technician, Finance Officer).
03	staff_type	ENUM	NOT NULL (Administrative Technical)	Distinguishes administrative from technical staff for routing.

Table 2.5: Staff Entity Attributes

2.2.5. Administrator (Subtype of User)

SRS References: 2.3.4, 3.4.1.2, 3.4.1.3 [1]

The Administrator is a single flat functional role representing a system-wide operator. All Administrators have full access to all departments, all complaints, all users, and all system configuration functions. No additional attributes beyond the shared primary key are required since the Administrator’s identity and department context are fully captured by the User supertype.

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	PK, NOT NULL	Shared PK enforces supertype–subtype relationship. Grants system-wide access to all departments and operations [1].

Table 2.6: Administrator Entity Attributes

2.2.6. Complaint Handler (Subtype of User)

SRS References: 2.3.4, 3.1.3.1 [1]

The Complaint_Handler is a department-scoped functional role. A handler is assigned to one or more trackers within a single department and is responsible for reviewing, updating, and resolving complaints routed through those trackers. The handler’s department is derived through the Tracker they handle and does not need to be stored directly on this table. Access level within the handler role is differentiated by handler_level as described by BR-01 [2].

#	Name	Data Type	Constraint	Description
01	user_id (PK)	INT	PK, NOT NULL	Shared PK enforces supertype–subtype relationship .
02	handler_level	ENUM	NOT NULL (Senior Junior)	Senior handlers have department-wide oversight and can manage junior handlers and configure trackers within their department. Junior handlers can only view and resolve complaints on their assigned tracker.

Table 2.7: Complaint_Handler Entity Attributes

2.2.7. Department

SRS References: 3.1.2.1 [1]

#	Name	Data Type	Constraint	Description
01	department_id (PK)	INT	NOT NULL, AUTO_INCREMENT	System-generated identifier.
02	name	VARCHAR(150)	NOT NULL, UNIQUE	Department name displayed on complaints and admin dashboards.
03	description	TEXT	NULLABLE	Optional description of department responsibilities.
04	is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Inactive departments are hidden from submission forms without losing historical data per BR-15.

Table 2.8: Department Entity Attributes

2.2.8. Tracker

SRS References: 3.1.2.2 [1]

#	Name	Data Type	Constraint	Description
01	tracker_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier; referenced by Complaint for classification.
02	name	VARCHAR(150)	NOT NULL	Category name displayed to users during complaint submission [1].
03	description	TEXT	NULLABLE	Guidance text helping users choose the correct tracker.

#	Name	Data Type	Constraint	Description
04	is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Inactive trackers are hidden from submission forms without losing historical links per BR-15.

Table 2.9: Tracker Entity Attributes

2.2.9. Complaint

SRS References: 3.1.2.3, 3.1.3.1, 3.1.3.3–3.1.3.7 [1]

#	Name	Data Type	Constraint	Description
01	complaint_id (PK)	INT	NOT NULL, AUTO_INCREMENT	System-generated ID displayed to users [1].
02	subject	VARCHAR(200)	NOT NULL, MAX 200 chars	Short title of the complaint per BR-16.
03	description	VARCHAR(2000)	NOT NULL, MAX 2000 chars	Detailed complaint body per BR-16.
04	priority	ENUM	NOT NULL (Low Medium High Critical)	Priority level used for filtering and handler assignment [1].
05	status	ENUM	NOT NULL (New Under Review In Progress Resolved Rejected Closed)	Complaint lifecycle states following the strict transition order defined in BR-06. Every transition is recorded in Status_History.
06	room_number	VARCHAR(5)	NULLABLE	Optional room or location reference provided by the complainant at submission per BR-16.
07	submitted_at	TIMESTAMP	NOT NULL	Submission timestamp for date-range filtering and resolution-time reporting [1].
08	resolved_at	TIMESTAMP	NULLABLE	Set when status changes to Resolved; enables resolution time analytics [1].
09	updated_at	TIMESTAMP	NOT NULL	Tracks last modification for audit purposes.

Table 2.10: Complaint Entity Attributes

2.2.10. Status History

SRS References: 3.1.3.1 [1]

#	Name	Data Type	Constraint	Description
01	history_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Surrogate key for the status change record.
02	old_status	ENUM	NOT NULL	Previous status; enables audit and transition reversal analysis [1].
03	new_status	ENUM	NOT NULL	Status the complaint moved to, constrained by the lifecycle order in BR-06.
04	changed_by	INT	NOT NULL	user_id of the Administrator or Complaint_Handler who performed the status change per BR-07.
05	changed_at	TIMESTAMP	NOT NULL	Exact timestamp of the transition for resolution-time analytics [1].
06	remarks	VARCHAR(500)	NULLABLE	Optional explanation for the status change (e.g., reason for rejection).

Table 2.11: Status_History Entity Attributes

2.2.11. Attachment

SRS References: 3.1.2.3 [1]

#	Name	Data Type	Constraint	Description
01	attachment_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for the file metadata record.
02	file_name	VARCHAR(255)	NOT NULL	Original filename displayed to administrators when reviewing evidence.
03	file_type	ENUM	NOT NULL (JPG PNG)	Only JPG and PNG permitted per BR-05 [1].
04	file_size_kb	INT	NOT NULL, MAX 2048	Maximum 2 MB per file per BR-05 [1].
05	storage_path	VARCHAR(512)	NOT NULL	Path or URL to retrieve the file from external object storage.
06	uploaded_at	TIMESTAMP	NOT NULL	Upload timestamp for ordering and audit.

Table 2.12: Attachment Entity Attributes

2.2.12. Comment

SRS References: 3.1.4.1 [1]

#	Name	Data Type	Constraint	Description
01	comment_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for the comment.
02	body	VARCHAR(1000)	NOT NULL, MAX 1000 chars	Comment content; 1000-character limit per BR-16 [1].
03	created_at	TIMESTAMP	NOT NULL	Timestamp for chronological display of the conversation.

Table 2.13: Comment Entity Attributes

2.2.13. Feedback

SRS References: 3.1.5.2 [1]

#	Name	Data Type	Constraint	Description
01	feedback_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for the feedback record.
02	overall_rating	TINYINT	NOT NULL, RANGE 1–5	Mandatory overall satisfaction score per BR-13 [1].
03	response_time_rating	TINYINT	NULLABLE, RANGE 1–5	Optional rating for response speed [1].
04	communication_rating	TINYINT	NULLABLE, RANGE 1–5	Optional rating for quality of communication during handling [1].
05	resolution_rating	TINYINT	NULLABLE, RANGE 1–5	Optional rating for quality of the actual resolution [1].
06	text_comment	TEXT	NULLABLE	Free-text elaboration accompanying the ratings per BR-16.
07	submitted_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Timestamp for ordering and trend analysis [1].

Table 2.14: Feedback Entity Attributes

2.2.14. Audit Log

SRS References: 3.4.1.2 [1]

#	Name	Data Type	Constraint	Description
01	log_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for the log entry.

#	Name	Data Type	Constraint	Description
02	action_type	ENUM	NOT NULL (Login Login_Failed Complaint_Submitted Status_Changed Comment_Added Complaint_Assigned Account_Locked Account_Unlocked Attachment_Uploaded Feedback_Submitted)	Enumerated action type enables filtering on specific event categories per BR-14 [1].
03	description	TEXT	NULLABLE	Human-readable context (e.g., “Status changed from New to In Progress by admin_id=7”).
04	ip_address	VARCHAR(45)	NULLABLE	Client IP recorded for security events such as login failures and account lockouts [1].
05	occurred_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Exact timestamp for accountability per BR-14 [1].

Table 2.15: Audit_Log Entity Attributes

2.3. BUSINESS RULES

The following business rules define constraints governing the behaviour of the CMTS database and application layer. Each rule is derived from the SRS [1] or the Database Design Proposal [2].

- **BR-01:** A user must belong to exactly one primary identity subtype: Student, Faculty, or Staff [1]. A user may additionally hold a functional role subtype of Administrator or Complaint_Handler simultaneously with their primary subtype. For example, a Faculty member may also be a Complaint_Handler. A user cannot hold both Administrator and Complaint_Handler roles at the same time. Every Complaint_Handler must have a handler_level of either Senior or Junior [2].
- **BR-02:** A user’s account is automatically locked after five consecutive failed login attempts [1]. Only an Administrator may unlock the account. The user cannot self-unlock. The failed_login_count is reset to zero upon a successful login.
- **BR-03:** Anonymous complaint submission is prohibited [1]. Every complaint must be linked to a registered, authenticated user account. The complainant_id in the Complaint table must never be NULL.
- **BR-04:** A complaint must be classified under exactly one tracker, and the tracker must belong to an active department at the time of submission [1].

- **BR-05:** Attachments may only be uploaded at the time of complaint submission. No attachments may be added after submission. A complaint may have a maximum of five attachments. Each must be in JPG or PNG format and must not exceed 2 MB in size [1].
- **BR-06:** The status of a complaint must follow the defined lifecycle in strict sequential order: *New* → *Under Review* → *In Progress* → *Resolved* → *Closed*. A complaint may be set to *Rejected* from any active status [1]. Every transition is recorded immutably in *Status_History*.
- **BR-07:** Only an Administrator or a Complaint_Handler may change the status of a complaint [1]. The *changed_by* field in *Status_History* must reference a user whose *is_administrator* or *is_complaint_handler* flag is TRUE.
- **BR-08:** A tracker and its assigned Complaint_Handler must both belong to the same single department [2]. A tracker cannot span multiple departments. A Complaint_Handler cannot be assigned to trackers from different departments simultaneously.
- **BR-09:** When an Administrator removes a Complaint_Handler from the system, all open complaints currently assigned to that handler's tracker must be reassigned to another active handler within the same department before the removal is committed [2]. Every active tracker must have at least one assigned handler at all times.
- **BR-10:** Comments on a complaint may only be posted by the original complainant, an Administrator, or a Complaint_Handler [1]. Since a non-admin user can only view their own complaints, they can only comment on complaints they filed themselves.
- **BR-11:** No comments may be posted on a complaint whose status is *Closed* or *Rejected*. Comment submission is prohibited once a complaint has reached a terminal status [2].
- **BR-12:** Feedback is automatically prompted when a complaint status changes to *Resolved* [1]. Feedback may only be submitted when the complaint status is *Resolved*. Feedback submission is prohibited on complaints in any other status including *Closed*.
- **BR-13:** Exactly one feedback record may be submitted per complaint [1]. The feedback must be submitted by the original complainant only. No other user may submit feedback on a complaint they did not file. Once submitted, feedback cannot be edited or deleted.
- **BR-14:** All records in the *Audit_Log* are insert-only [1]. No application role is granted UPDATE or DELETE permissions on the audit log table. *Status_History* records are likewise insert-only and must never be modified or deleted after creation.
- **BR-15:** Inactive departments and trackers are hidden from complaint submission forms but are retained in the database to preserve historical complaint records [1]. Historical complaints linked to inactive departments or trackers must remain fully accessible.
- **BR-16:** The following field length limits are enforced at both the application and database layers [1]: complaint subject 200 characters; complaint description 2,000 characters; comment body 1,000 characters; feedback text 500 characters; room number 5 characters.

2.4. ENTITY RELATIONSHIP DIAGRAM

2.4.1. Relationships, Cardinality, and Multiplicity

The CMTS schema comprises 18 relationships. The table below documents each relationship's type, cardinality, participation constraints, and originating SRS requirement. Cardinality notation follows the crow's foot convention, originally introduced by Gordon Everest [3].

ID	Relationship	Card.	SRS Ref	Multiplicity & Participation
R1	User is a Student	1:1	2.3.1 [1]	User mandatory, Student mandatory. Shared primary key enforces the supertype–subtype link.
R2	User is a Faculty	1:1	2.3.2 [1]	User mandatory, Faculty mandatory. Shared primary key enforces the supertype–subtype link.
R3	User is a Staff	1:1	2.3.3 [1]	User mandatory, Staff mandatory. Shared primary key enforces the supertype–subtype link .
R4	User is an Administrator	1:1	2.3.4 [1]	User mandatory, Administrator mandatory. A user holds this role in addition to a primary subtype per BR-01.
R5	User is a Complaint_Handler	1:1	2.3.4 [1]	User mandatory, Complaint_Handler mandatory. A user holds this role in addition to a primary subtype per BR-01.
R6	Department has Tracker	1:M	3.1.2.1 [1]	Department optional (zero or more trackers), Tracker mandatory (must belong to one department).
R7	User belongs to Department	M:1	3.1.1.1 [1]	Department optional (zero or more users), User mandatory (must belong to one department).
R8	User submits Complaint	1:M	3.1.2.3 [1]	User optional (may have zero complaints), Complaint mandatory (complainant_id NOT NULL per BR-03).

ID	Relationship	Card.	SRS Ref	Multiplicity & Participation
R9	Complaint_Handler handles Tracker	1:M	3.1.3.1 [1]	Complaint_Handler optional, Tracker mandatory (must have one handler). One handler manages many trackers of the same department per BR-08.
R10	Tracker classifies Complaint	1:M	3.1.2.2 [1]	Tracker optional (zero or more complaints), Complaint mandatory (must be classified under one active tracker per BR-04).
R11	Complaint has Status_History	1:M	3.1.3.1 [1]	Complaint mandatory (at least one history record on creation), Status_History mandatory (must belong to a complaint).
R12	Complaint has Attachment	1:M	3.1.2.3 [1]	Complaint optional (zero to five attachments), Attachment mandatory.
R13	Complaint receives Comment	1:M	3.1.4.1 [1]	Complaint optional (zero or more comments), Comment mandatory.
R14	User writes Comment	1:M	3.1.4.1 [1]	User optional (writes zero or many comments), Comment mandatory.
R15	Complaint receives Feedback	1:1	3.1.5.2 [1]	Complaint optional (zero or one feedback), Feedback mandatory. One feedback per complaint only per BR-13.
R16	User generates Audit_Log	1:M	3.4.1.2 [1]	User optional, Audit_Log mandatory.
R17	Complaint recorded in Audit_Log	1:M	3.4.1.2 [1]	Mandatory on both sides.
R18	Administrator updates Status_History	1:M	3.1.3.1 [1]	Administrator optional, Status_History mandatory.

Table 2.16: CMTS Relationship Summary

2.4.2. EER Diagram

[Click here to view diagram](#)

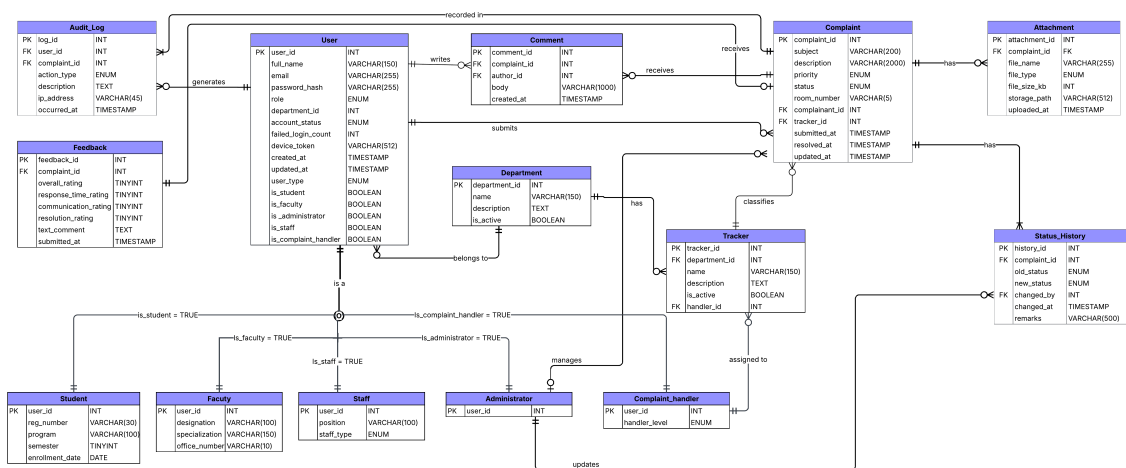


Figure 2.1: CMTS Entity Relationship Diagram

BIBLIOGRAPHY

- [1] Complaint Management and Tracking System (CMTS), “Software Requirements Specification (SRS),” Namal University, Mianwali, Pakistan, Dec. 2025.
- [2] Complaint Management and Tracking System (CMTS), “Database Design Proposal,” Namal University, Mianwali, Pakistan, Mar. 2026.
- [3] G. C. Everest, “Basic Data Structure Models Explained with a Common Example,” in *Proc. 5th Texas Conf. Computing Systems*, Austin, TX, USA, 1976, pp. 39–46.

APPENDIX: AI USAGE AND PROMPTS

The following table documents the use of AI tools during the development of this project.

#	Prompt Used	Purpose	Tool Used
01	“Analyze the given SRS document and then identify missing entities and attributes required to model the database of the system.”	Entity Identification	Claude (Anthropic)
02	“Convert the attached CMTS document into a latex database design report following the uploaded template.”	LaTeX Report Generation	Claude (Anthropic)
03	“Analyze the given SRS and EERD diagram, then point out flaws and missing elements in EERD.”	Error Detection	Claude (Anthropic)
04	“List business rules that are missing or anything that is ambiguous (attached document).”	Business Rule Analysis	Claude (Anthropic)
05	“Go through the provided template and then act as strict teacher whose goal is to give as less marks as possible and point of mistakes in formatting and other issues (attached document).”	Correct Formatting	Claude (Anthropic)

Table 2.17: AI Tool Usage